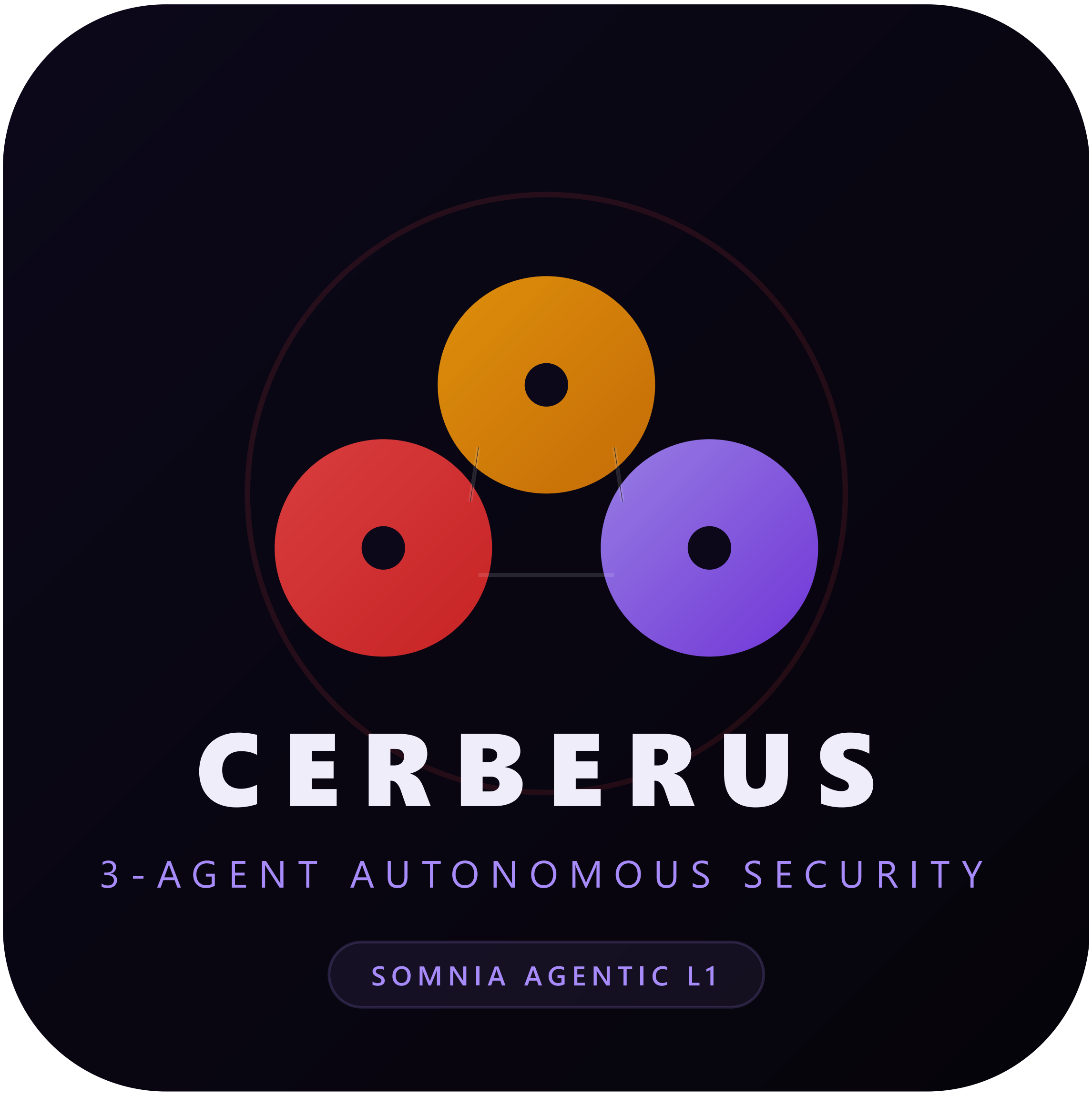
Cerberus — 3-Agent Autonomous Security

Somnia Agentathon 2026



The Problem

Smart contracts have no immune system.

- Oracles get manipulated — nobody notices
- Price feeds diverge — funds drain before humans react
- Exploits happen in seconds, responses take hours

> *"Who watches the watchers?"*

The Solution:

Three AI agents. One 60-second heartbeat. Zero trust.

```
📡 OracleGuard      → detects anomalies in 3 price feeds
🧠 ThreatClassifier → deterministic LLM classifies threat
⚡ CircuitBreaker   → auto-pauses compromised contracts
```

Every decision = verifiable receipt on Somnia Agentic L1.

Architecture

```
graph LR
    A[CoinGecko] --> OG
    B[Binance] --> OG
    C[CryptoCompare] --> OG
    OG[OracleGuard] -->|anomaly| TC[ThreatClassifier]
    TC -->|CRITICAL| CB[CircuitBreaker]
    CB -->|pause| SC[CerberusSentinel.sol]
    OG -->|receipt| SL[Somnia L1]
    TC -->|receipt| SL
    CB -->|receipt| SL
```

Somnia primitives used: JSON API Request + LLM Inference + Verifiable Receipts

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Live Dashboard

- **Overview** — 6 metric cards, ETH/USD price, pipeline status
- **Agent Pipeline** — 3 agent detail cards with Somnia primitive badges
- **Event Log** — color-coded, filterable (Critical/Medium/OracleGuard/Errors)
- **About** — wallet, contract, chain, RPC info

One-click anomaly simulation for demo. Auto-refresh 2s.

Why This Wins

Criterion	How Cerberus Delivers
Agent-First	Agents discover, classify, act — no human in loop
Innovation	Deterministic LLM for security classification (first of its kind)
Autonomous	60s heartbeat, self-healing, runs forever
Functionality	Working prototype, 3 live price feeds, deployed on Railway

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Links

- **GitHub:** https://github.com/Gideon145/cerberus

- **Live Demo:** https://cerberus-production-8429.up.railway.app
- **CV:** Opukeme-Gideon-Somnia.pdf

Built in 3 days by a solo developer for the Somnia Agentathon.